

Exercise 7: Neural Tangent Kernel

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Problem 1 (NTK for two-layer ReLU network):

Consider a two-layer neural network with the second layer fixed,

$$f(\mathbf{W}, \mathbf{x}) = \frac{1}{\sqrt{m}} \sum_{r=1}^m \sigma(\mathbf{w}_r^\top \mathbf{x})$$

where $\mathbf{x} \in \mathbb{R}^d$ is the input, $\mathbf{W} = (\mathbf{w}_r)_{r=1}^m \in \mathbb{R}^{m \times d}$ is a matrix containing the weight vectors $\mathbf{w}_r \in \mathbb{R}^d$ of the first layer, and σ is the ReLU activation function. Throughout this problem, only the first layer (with weight vectors $\mathbf{w}_r$, $r = 1, \dots, m$) is trained. We are given a training dataset $(\mathbf{x}_i, y_i)$ where each $\mathbf{x}_i \in \mathbb{R}^d$ is a feature vector such that $\|\mathbf{x}_i\| = 1$, and $y_i \in \mathbb{R}$ is the corresponding target label. We consider the following square loss:

$$\ell(\mathbf{W}) = \frac{1}{2} \sum_{i=1}^n (f(\mathbf{W}, \mathbf{x}_i) - y_i)^2.$$

We optimize over each $\mathbf{w}_r$ using gradient flow:

$$\frac{d\mathbf{w}_r(t)}{dt} = -\frac{\partial \ell(\mathbf{W}(t))}{\partial \mathbf{w}_r(t)} \in \mathbb{R}^d$$

for $r = 1, \dots, m$.a) We denote $u_i(t) = f(\mathbf{W}(t), \mathbf{x}_i)$ the prediction on input $\mathbf{x}_i$ at time t . Show that, at any time t , we have

$$\frac{\partial \ell(\mathbf{W}(t))}{\partial \mathbf{w}_r(t)} = \sum_{i=1}^n (u_i(t) - y_i) \frac{\partial f(\mathbf{W}(t), \mathbf{x}_i)}{\partial \mathbf{w}_r(t)} \in \mathbb{R}^d$$

b) Let $\mathbf{u}(t) = (u_1(t), \dots, u_n(t)) \in \mathbb{R}^n$ be the prediction vector at time t . Show that, using chain rule on $\mathbf{u}(t)$, the dynamics of the predictions can be written as

$$\frac{d}{dt} \mathbf{u}(t) = \mathbf{H}(t)(\mathbf{y} - \mathbf{u}(t)),$$

where $\mathbf{H}(t)$ is an $n \times n$ matrix with (i, j) -th entry

$$\mathbf{H}_{ij}(t) = \sum_{r=1}^m \left\langle \frac{\partial f(\mathbf{W}(t), \mathbf{x}_i)}{\partial \mathbf{w}_r(t)}, \frac{\partial f(\mathbf{W}(t), \mathbf{x}_j)}{\partial \mathbf{w}_r(t)} \right\rangle$$

Note that the so-called Gram matrix $\mathbf{H}$ defined above is essentially the neural tangent kernel on the training data.c) Show that, at any time t , we have the following expression of the entries in the Gram matrix $\mathbf{H}$:

$$\mathbf{H}_{ij}(t) = \frac{1}{m} \sum_{r=1}^m \mathbf{x}_i^\top \mathbf{x}_j \mathbb{I} \{ \mathbf{w}_r(t)^\top \mathbf{x}_i \geq 0, \mathbf{w}_r(t)^\top \mathbf{x}_j \geq 0 \}$$

where the indicator $\mathbb{I}\{A\}$ is 1 when the constraint A holds; 0 otherwise.Hint: The derivative $\sigma'(z)$ of the ReLU function is the step function $\text{step}(z) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$.d) If we assume the weight vectors $\mathbf{w}_r$ are initialized as independent standard Gaussian vectors, i.e. $\mathbf{w}_r(0) \sim \mathcal{N}(0, \mathbf{I})$ for $r = 1, \dots, m$, and we take the limit of the layer width $m \rightarrow \infty$, we have the so-called neural tangent kernel (NTK) matrix $\mathbf{H}^\infty$ with entries:

$$\mathbf{H}_{ij}^\infty = \lim_{m \rightarrow \infty} \mathbf{H}_{ij}(0) = \mathbb{E}_{\mathbf{w} \sim \mathcal{N}(0, \mathbf{I})} [\mathbf{x}_i^\top \mathbf{x}_j \mathbb{I} \{ \mathbf{w}^\top \mathbf{x}_i \geq 0, \mathbf{w}^\top \mathbf{x}_j \geq 0 \}]$$

Show that we have the following closed-form for the entries of NTK $\mathbf{H}^\infty$:

$$\mathbf{H}_{ij}^\infty = \frac{\mathbf{x}_i^\top \mathbf{x}_j (\pi - \arccos(\mathbf{x}_i^\top \mathbf{x}_j))}{2\pi}, \quad \forall i, j = 1, \dots, n. \quad (1)$$

Problem 2 (Upper bound of classification error):

Assume we have a dataset with n data points, $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where each input $\mathbf{x}_i \in \mathbb{R}^d$ has norm 1 and $y_i = \beta^\top \mathbf{x}_i$ for some $\beta \in \mathbb{R}^d$. Write $\mathbf{X} = (\mathbf{x}_i)_{i=1}^n \in \mathbb{R}^{n \times d}$.

We consider the same two-layer ReLU network used in the previous problem. An upper bound on the classification error of the NTK is given by

$$\frac{\sqrt{2\mathbf{y}^\top (\mathbf{H}^\infty)^{-1} \mathbf{y} \cdot \text{tr}(\mathbf{H}^\infty)}}{n},$$

where the NTK matrix $\mathbf{H}^\infty$ is defined in Eq. (1).

This complexity measure can be derived using *Rademacher complexity* bounds, which will be the subject of a later course. Our goal will be to derive a bound on this complexity measure that decreases as n increases. This will imply that the NTK can achieve zero classification error given a sufficient large number of datapoints n .

a) First, show that $\mathbf{H}^\infty$ admits the following expression of entries:

$$\mathbf{H}_{ij}^\infty = \frac{\mathbf{x}_i^\top \mathbf{x}_j}{4} + \frac{1}{2\pi} \sum_{l=0}^{\infty} \frac{(2l)!}{2^{2l}(l!)^2} \frac{(\mathbf{x}_i^\top \mathbf{x}_j)^{2l+2}}{2l+1}.$$

Hint: Use the following Taylor approximation of $\arccos(z)$:

$$\arccos(z) = \frac{\pi}{2} - \sum_{l=0}^{\infty} \frac{(2l)!}{2^{2l}(l!)^2} \frac{z^{2l+1}}{2l+1}.$$

b) Using the following facts: (You do not need to prove them)

- $4\mathbf{K}^{-1} - (\mathbf{H}^\infty)^{-1}$ is a positive semi-definite matrix; (why?)
- $\text{tr}(\mathbf{H}^\infty) \leq n$; (why?)
- the operator norm of the matrix $\mathbf{X}(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{X}^\top$ is less than or equal to 1. (why?)

and prove the following upper bound on the classification error of the NTK :

$$\frac{\sqrt{2\mathbf{y}^\top (\mathbf{H}^\infty)^{-1} \mathbf{y} \cdot \text{tr}(\mathbf{H}^\infty)}}{n} \leq \frac{2\sqrt{2}\|\beta\|_2}{\sqrt{n}}.$$